

Mike Hall

Principal Software Engineer

Chicago, IL • mike@just3ws.com • (847) 877-3825 • just3ws.com • linkedin.com/in/just3ws/ • github.com/just3ws

TECHNICAL LEADERSHIP & PRODUCTION SYSTEMS MODERNIZATION

Principal Software Engineer specializing in high-consequence legacy modernization, distributed systems architecture, and platform resilience. Combines 20+ years across software engineering and platform architecture with OpenTelemetry distributed tracing, cross-lane boundary mediation, and zero-downtime data migrations across Ruby on Rails, PostgreSQL, and cloud infrastructure.

AREAS OF FOCUS

- 01 Architecture & Modernization
- 02 Observability & AI-Augmented Engineering

EXPERIENCE

Associate Director, Staff Engineer

OneMain Financial | Remote | January 2021 - February 2026

Served as Senior Technical IC and Software Architecture Lead for the Acquisition lane of a regulated financial enterprise, establishing system cartography and observability across the end-to-end loan application and origination lifecycle, while guiding legacy modernization, cybersecurity investigations, and cross-team platform reliability.

- Technical Leadership & Team Consolidation: Progressed from Digital Origination Team Lead to Software Architect for the Acquisition Lane, merging two attrition-impacted teams into a unified, high-functioning engineering unit before converting to Associate Director, Staff Engineer upon corporate architecture track restructuring.
- ACQ Enablement & Funnel State Mapping: Founded and technically led the ACQ Enablement team to establish systemic cartography across customer journeys: inventorying the intersection of business decisions, customer actions, system state transitions, and source code execution paths. Built high-fidelity funnel lifecycle signals (entry, exit, error) with complete runtime attribution (A/B test variants, feature flags, configuration states) to accelerate cybersecurity investigations and create safety gates for continuous legacy modernization.
- Observability & Enterprise Origination Trace: Founded and led the OpenTelemetry Working Group (OTel WG) to establish the Enterprise Trace across the end-to-end loan application and digital origination lifecycle, connecting a complex Rails distributed monolith, MuleSoft, and IBM mainframe backends. Leveraged distributed telemetry and state mapping to diagnose and resolve a persistent multi-service defect impacting 4% of traffic, eliminating silent application loss at final underwriting and e-sign stages.
- AI-Augmented Systems Investigation: Introduced privacy-conscious local AI workflows for legacy code analysis and engineering support, expanding investigative capacity while keeping human review, security constraints, and source evidence explicit.

Senior Backend Developer

SK Holdings, Inc. | Chicago, IL | January 2019 - December 2020

Led backend stability, performance, and modernization for high-traffic Rails products, improving full-text search, campaign delivery pipelines, and data layer reliability while systems remained under production load.

- Search Modernization & Infrastructure De-Risking: Eliminated external cluster failure modes and lowered operating footprint by transitioning legacy Sphinx infrastructure to optimized MySQL full-text search and relational indices under live production traffic.
- Zero-Downtime Platform Upgrades: Led Ruby and Rails framework upgrades across multiple core applications, sequencing database migrations and dependency updates to maintain uptime during platform modernization.

Senior Software Developer

ActiveCampaign | Chicago, IL | September 2018 - December 2018

Short-term senior contract on the Contacts team of a large-scale PHP/MySQL CRM platform. Focused on testability, performance, and change safety inside a tightly coupled legacy backend while extending the Ember.js frontend test suite.

- Testable Query Patterns: Replaced ad-hoc global data access functions with structured, cacheable query patterns in the PHP backend, improving performance and enabling reliable automated testing where none existed before.
- Legacy Boundary Isolation: Identified and isolated structural constraints in the core contact management subsystem, reducing risk for incremental change without requiring large-scale rewrites.
- Frontend Test Hardening & Boundary Verification: Hardened Ember.js frontend test coverage across core contact management workflows, creating verification gates that allowed safe refactoring of legacy PHP data paths without regression.
- Reusable Pagination Constructs: Refactored shared Ember.js pagination and interaction logic into reusable mixins, eliminating duplication across contact list views.

SELECTED CURRENT WORK

Principal Architect

Local AI Orchestration & Developer Runtime | Independent Project | 2026 - Present

Built and operate three MCP (Model Context Protocol) servers that expose live system state as callable tools to any MCP client, alongside the Claude Code skills, context-isolated subagents, and commit-time checks that keep agent work bounded and reviewable.

- Bidirectional Knowledge Server: Built ctx-mcp, an MCP server fronting a PostgreSQL knowledge suite that agents both query and write back to, so methodologies and lessons learned in one session are retrievable in the next rather than re-derived.
- Telemetry-Driven Diagnosis: Built o2-mcp, exposing on-demand OpenObserve telemetry (errors, slow spans, failed jobs, distributed traces) as agent-callable tools, so an agent diagnoses a running system from observed behavior instead of inferring from source.
- Context as a Budgeted Resource: Designed context-isolated subagents that keep high-volume audit work out of the main conversation's context window, with per-client registration tooling and pre-commit hooks that reject any agent asset that isn't self-describing.

Principal Architect

Phalanx Duel | Independent Project | 2022 - Present

Designing and building a real-time tactical game platform as a current hands-on laboratory for deterministic systems, product architecture, and controlled AI-assisted engineering.

- Designed a deterministic engine with replayable action logs and server-authoritative state transitions, making complex outcomes reproducible and independently verifiable.
- Built replay validation, adversarial coverage, and CI verification gates so state transitions can be checked across the complete lifecycle.
- Uses controlled agent workflows with bounded tasks, acceptance criteria, automated checks, and human-reviewed outputs rather than treating model output as authority.

Principal Architect

WWWorkRemote | Independent Project | 2021 - Present

Rails 8 platform for multi-source job ingestion, semantic matching, and application automation, built as a working laboratory for local-first LLM orchestration and treating scraped third-party text as hostile input.

- Data Ingestion & Normalization: Built a multi-source Rails ingestion pipeline and PostgreSQL storage schema, unifying fragmented job market datasets for structured downstream analysis.
- Semantic Search & Matching: Implemented vector search embeddings via pgvector, enabling high-precision semantic matching between complex candidate profiles and job requirements.
- Local-First AI Orchestration: Designed a provider-agnostic LLM orchestration layer behind a YAML-driven model registry, local-first by default (llama.cpp/Ollama primary, hosted Claude as fallback), so routine inference never leaves the machine while a stronger hosted model can be selected per call site.
- Prompt Injection Defense: Built a four-stage guardrails pipeline (normalize, weighted heuristic scan, risk classify, output validate) screening every untrusted scraped document before it reaches a model, scoring known injection patterns plus an instruction-density heuristic to secure ingestion against hostile inputs.
- Browser Extension & Lifecycle Capture: Built an MV3 Chrome extension with 16 provider adapters that extracts postings from live ATS pages and captures the full application lifecycle (questions asked and answers given) back into the platform.
- Static Signal Integration: Injected static analysis rules and architectural constraints into model context windows, preventing drift between generated outputs and system security contracts.

CORE CAPABILITIES & TECHNOLOGIES

ARCHITECTURE & MODERNIZATION

- Legacy Modernization & Architecture Discovery,
- Zero-Downtime Data Migration & Integrity,
- Incident Response & Root Cause Analysis,
- Distributed Systems & Asynchronous Processing

OBSERVABILITY & AI-AUGMENTED ENGINEERING

- OpenTelemetry & Observability Architecture,
- Model Context Protocol (MCP) Server Design,
- LLM Orchestration & Deterministic Evaluation,
- Platform Enablement & Developer Velocity

TECHNOLOGIES

- Ruby on Rails, PostgreSQL, Sidekiq & Redis, AWS, Kubernetes & Docker, CI/CD

SELECTED EARLIER EXPERIENCE

- Senior Software Developer, BenchPrep | March 2017 - February 2018 | Chicago, IL : Owned enterprise assessment workflows, leading correctness and platform security in a high-concurrency environment.
- Senior Software Developer, ReachLocal | March 2015 - November 2016 | Remote : Owned API design and modernization strategy, leading incremental legacy migration for a high-volume digital marketing platform.
- Open-Source Transition Lead, Coderwall | January 2014 - December 2014 | Remote : Hired as a contractor by the founder to lead the open-source transition of the Coderwall developer reputation platform, a Y Combinator-backed professional network for software engineers (856 GitHub stars, 304 forks). Delivered security hardening, proprietary service extraction, infrastructure modernization, and community leadership as the top contributor to the open-source codebase.
- Software Engineer & Technical Onboarding Lead (Fraud & Taxonomy Systems), Groupon | July 2011 - May 2013 | Chicago, IL : Owned core backend systems for fraud and taxonomy, leading global engineering enablement during hyper-growth.

CAREER FOUNDATIONS

1999 - 2011

Built and operated enterprise integrations, transactional systems, and early web products across consulting and product organizations, establishing the production discipline and community-centered craftsmanship that still shape my leadership.

- **Revenue-critical commerce:** Designed real-time inventory, locking, and fulfillment services for a high-volume ticket marketplace later acquired by a major industry operator.
- **Consulting and software craftsmanship:** Delivered systems across client environments while mentoring engineers, adopting Ruby and Rails, and helping build Chicago's craftsmanship community.
- **Enterprise and operational systems:** Built integration and operational software across large organizations, grounded in .NET, SQL, messaging, production support, and direct user needs.